

Counterexamples to Gao's zero-sum invariant conjecture

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Abstract

For every odd integer $n \geq 3$, we prove that $\nu(C_2^3 \oplus C_{2n}) = 2n + 2 = \mathbf{d}(C_2^3 \oplus C_{2n})$. This disproves Gao's conjecture that $\nu(G) = \mathbf{d}(G) - 1$ for every nontrivial finite abelian group. The construction is a zero-sum-free sequence of length $2n + 1$ with exactly six omitted nonzero subsequence sums. These sums form the product of the three nonzero elements of an elementary two-group of rank two with two adjacent cyclic residues. Their differences force every coset containing them to contain zero. We also give explicit extremal extensions and show that the index-two refinement and the associated local invariant attain their larger possible values.

1 The conjecture and the counterexamples

Let G be a nontrivial finite abelian group, written additively. A sequence over G is a finite multiset of elements, written multiplicatively. Thus g^k denotes k occurrences of g . For a sequence $T = g_1 \cdots g_\ell$, put

$$\Sigma(T) = \left\{ \sum_{i \in I} g_i : \emptyset \neq I \subseteq [1, \ell] \right\}, \quad \Sigma^*(T) = \Sigma(T) \cup \{0\}.$$

The sequence is zero-sum free if $0 \notin \Sigma(T)$. Its set of holes is

$$R(T) = G \setminus \Sigma^*(T).$$

Write $\mathbf{d}(G)$ for the greatest length of a zero-sum-free sequence and $\mathbf{D}(G) = \mathbf{d}(G) + 1$ for the Davenport constant.

The invariant $\nu(G)$ is the least integer $\ell \geq 0$ such that every zero-sum-free sequence T of length at least ℓ satisfies

$$R(T) \subseteq \alpha + H \quad \text{for some } H \subsetneq G \text{ and } \alpha \in G \setminus H. \tag{1}$$

The exclusion of zero from the coset is part of the definition. Merely putting the holes in a proper subgroup does not establish (1).

Gao conjectured that $\nu(G) = \mathbf{d}(G) - 1$; equivalently, $\mathbf{D}(G) = \nu(G) + 2$. The latter formulation appears in Gao and Thangadurai [3, Definition 3 and Conjecture 4]. Geroldinger and Yang [4, Introduction and Section 6] restate it as an open conjecture. Their recent results include the family $C_2^2 \oplus C_{2n}$ and certain groups with four elementary two-group factors. We give counterexamples with three such factors.

Theorem 1.1. *Let $n \geq 3$ be odd, and let*

$$G = \langle e_1 \rangle \oplus \langle e_2 \rangle \oplus \langle e_3 \rangle \oplus \langle f \rangle, \quad \text{ord}(e_i) = 2, \quad \text{ord}(f) = 2n.$$

Set

$$q = \frac{n+1}{2}, \quad C = \{e_1, e_2, e_3, e_1 + e_2 + e_3\}, \quad W = \langle e_1 + e_2, e_1 + e_3 \rangle.$$

Then the sequence

$$T_n = f^{2n-3} \prod_{c \in C} (c + qf) \quad (2)$$

is zero-sum free, has length $\mathbf{d}(G) - 1 = 2n + 1$, and has exactly the six holes

$$R(T_n) = (W \setminus \{0\}) + \{(n-1)f, nf\}. \quad (3)$$

No coset excluding zero contains these holes. Consequently,

$$\nu(C_2^3 \oplus C_{2n}) = \mathbf{d}(C_2^3 \oplus C_{2n}) = 2n + 2. \quad (4)$$

The Davenport formula used here is classical. Baayen [1, p. 1] proves that every sequence of length $2n + 3$ in this group has a nonempty zero-sum subsequence for odd n . On the other hand, $e_1 e_2 e_3 f^{2n-1}$ is zero-sum free. Thus $\mathbf{d}(G) = 2n + 2$. The same formula is recorded in [4, Proposition 2.1]; it is not a new result of this paper.

2 A coset obstruction and the exact holes

We start with the obstruction that makes the construction effective.

Lemma 2.1. *Let u, v be distinct nonzero elements of order two in an abelian group, let f be any group element, and let $b \in \mathbb{Z}$. No coset $\alpha + H$ with $\alpha \notin H$ contains*

$$\{u, v, u + v\} + \{bf, (b+1)f\}.$$

Proof. Suppose all displayed elements lie in $\alpha + H$. The difference of $u + (b+1)f$ and $u + bf$ gives $f \in H$. The difference of $u + bf$ and $v + bf$ gives $u - v = u + v \in H$. Therefore $u + v + bf \in H$. This element also lies in $\alpha + H$, forcing $\alpha \in H$, a contradiction. $\square$

Proof of Theorem 1.1. Put $E = \langle e_1, e_2, e_3 \rangle$. The four elements of C form a circuit in E : their sum is zero, and no proper nonempty subset has sum zero. Also,

$$4q = 2n + 2 \equiv 2 \pmod{2n}. \quad (5)$$

Any zero-sum subsequence of T_n must therefore use none or all four of the terms $c + qf$. In the former case it is f^k with $1 \leq k \leq 2n-3$, and has nonzero sum. In the latter case its sum is $(k+4q)f = (k+2)f$ with $0 \leq k \leq 2n-3$, again nonzero. Thus T_n is zero-sum free, and its length is $2n + 1$.

We determine all its holes, including those that will not be needed to disprove the conjecture. Enumerate $C = \{c_1, c_2, c_3, c_4\}$. The map

$$\{0, 1\}^4 \longrightarrow E, \quad (\epsilon_i) \longmapsto \sum_i \epsilon_i c_i$$

is a surjective homomorphism of elementary two-groups with kernel consisting of the zero vector and the all-one vector. Hence each $w \in E$ is represented by precisely two subsets of C , which are complements. Let their sizes be k and $4 - k$.

Selecting either subset and between zero and $2n - 3$ copies of f shows that the cyclic coordinates of $\Sigma^*(T_n)$ above w are

$$(kq + [0, 2n - 3]) \cup ((4 - k)q + [0, 2n - 3]) \quad \text{in } \mathbb{Z}/(2n)\mathbb{Z}.$$

Consequently the omitted coordinates above w form the intersection

$$\begin{aligned} & \{kq - 2, kq - 1\} \cap \{(4 - k)q - 2, (4 - k)q - 1\} \\ &= \{kq - 2, kq - 1\} \cap \{-kq, 1 - kq\}, \end{aligned} \tag{6}$$

where (5) was used.

Matching the first element of one pair with the second of the other would require $2kq \equiv 1$ or $3 \pmod{2n}$, which is impossible. The other two matches occur together, precisely when

$$2kq \equiv 2 \pmod{2n}.$$

Since $\gcd(q, n) = 1$ and $2q \equiv 1 \pmod{n}$, this is equivalent to $k \equiv 2 \pmod{n}$. For odd $n \geq 3$ and $0 \leq k \leq 4$, the only possibility is $k = 2$. The sums of pairs from C are exactly

$$e_1 + e_2, \quad e_1 + e_3, \quad e_2 + e_3,$$

each represented by complementary pairs. When $k = 2$, the two omitted coordinates in (6) are

$$2q - 2 = n - 1, \quad 2q - 1 = n.$$

This proves (3). Its six elements are distinct and nonzero.

Apply Lemma 2.1 with $u = e_1 + e_2$, $v = e_1 + e_3$, and $b = n - 1$. Thus T_n , of length $\mathbf{d}(G) - 1$, violates (1), so $\nu(G) \geq \mathbf{d}(G)$. For completeness, every zero-sum-free sequence S of length $\mathbf{d}(G)$ has $R(S) = \emptyset$: if $x \in R(S)$, then $(-x)S$ would be zero-sum free of greater length. The empty set satisfies (1), so $\nu(G) \leq \mathbf{d}(G)$. This proves (4). $\square$

3 Extremal extensions and the parity of the construction

Geroldinger and Yang define $\nu_p(G)$ by requiring the subgroup in (1) to have index p , for a prime divisor p of $|G|$. They also define a local invariant $\nu(S)$ for a maximal zero-sum-free sequence S : it is the least threshold for which all subsequences of S beyond that threshold have their holes in a coset excluding zero. Here maximal means that no additional term can be adjoined while retaining zero-sum freeness.

Corollary 3.1. *For the group in Theorem 1.1, we have $\nu_2(G) = \mathbf{d}(G)$. Let $u \in W \setminus \{0\}$. Then*

$$S_n = T_n(u + nf)$$

is zero-sum free of length $\mathbf{d}(G)$ and satisfies $\nu(S_n) = |S_n|$. Moreover,

$$U_n = T_n(u + nf)(u + (n + 1)f)$$

is a minimal zero-sum sequence of length $\mathbf{D}(G)$.

Proof. The inequality $\nu_2(G) \geq \nu(G) = \mathbf{d}(G)$ follows from the definitions, and the reverse inequality follows from the empty-hole property at length $\mathbf{d}(G)$. Since $u + nf$ is a hole of T_n and equals its negative, S_n is zero-sum free. It has greatest possible length, hence is maximal and has no holes. Its subsequence T_n violates the coset property, so the local threshold is exactly $|S_n|$.

The sum of T_n is $-f$, by (5). Therefore the sum of S_n is $u + (n - 1)f$, whose negative is $u + (n + 1)f$. Adjoining this term gives a zero-sum sequence. A proper zero-sum subsequence either lies in S_n , or has a nonempty zero-sum complement in S_n ; both are impossible. Thus U_n is minimal. $\square$

The parity restriction in the theorem is intrinsic to this particular sparse construction, rather than an omitted even case.

Proposition 3.2. *Let $n \geq 2$, let $G = C_2^3 \oplus \langle f \rangle$ with $\text{ord}(f) = 2n$, and use the same circuit $C \subseteq C_2^3$. For $t \in \mathbb{Z}$, the sequence*

$$f^{2n-3} \prod_{c \in C} (c + tf)$$

is zero-sum free if and only if $4t \equiv 2 \pmod{2n}$. Such a t exists if and only if n is odd.

Proof. A zero-sum subsequence must use either none or all four circuit terms. The former possibility is excluded by the number of copies of f . In the latter case the available cyclic sums are $4t + [0, 2n - 3]$. This interval modulo $2n$ excludes zero exactly when $4t$ is congruent to one or two. The first possibility is impossible by parity, giving the stated criterion. The congruence is soluble exactly when $\gcd(4, 2n)$ divides two, or equivalently when n is odd. $\square$

4 The smallest example and remaining questions

For $n = 3$, the counterexample lies in a group of order 48 and has the particularly simple form

$$T_3 = f^3(e_1 + 2f)(e_2 + 2f)(e_3 + 2f)(e_1 + e_2 + e_3 + 2f).$$

Its 127 nonempty subsequences have 41 distinct sums, all nonzero. The six holes are

$$\{e_1 + e_2, e_1 + e_3, e_2 + e_3\} + \{2f, 3f\}.$$

They lie in the proper subgroup $W \oplus \langle f \rangle$, but in no coset that excludes zero. This distinction explains why containment in a proper subgroup would not suffice in the conjecture.

The proof above is uniform in n and does not depend on finite computation. The accompanying verification checks the exact hole set, the difference-generated subgroup obstruction, the extremal extension, and the longest atom for every odd n from three through 101. It also separately enumerates all nonempty subsequences in the smallest example.

The theorem disproves the universal conjecture, but it does not classify all finite abelian groups with $\nu(G) = d(G)$. In particular, Proposition 3.2 only rules out this construction for even n ; it does not determine $\nu(C_2^3 \oplus C_{2n})$ in all those cases. A revised classification must distinguish the established positive families from the rank-four family exhibited here. The explicit six-hole configuration provides a local obstruction that such a classification must accommodate.

References

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